

CYBERSENTINEL

Behavioral Network Intrusion Detection & Response Prototype

Technical Implementation Report

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Ismael Abdallah Baby

Cybersecurity Student & Network Security Enthusiast

github.com/ismael-cybersec

1. Introduction

CyberSentinel is a custom-built network intrusion detection prototype developed to understand and implement real-time traffic monitoring and behavioral threat detection.

The goal of this project is to design a lightweight detection engine capable of analyzing live network traffic, identifying abnormal behaviors such as ICMP flood and SYN scanning, and generating structured security alerts.

This project was developed in Python using Scapy for packet capture and focuses on behavioral threshold-based detection rather than signature-only approaches.

2. Project Objectives

The primary objective of CyberSentinel is to design and implement a lightweight network detection prototype capable of monitoring live traffic and identifying abnormal behaviors in real time.

This project focuses on:

- Understanding packet-level network traffic analysis
- Implementing behavioral detection mechanisms
- Detecting common network attacks such as ICMP flooding and SYN scanning
- Generating structured security alerts
- Developing a modular and extensible detection architecture

Unlike traditional signature-based IDS systems, CyberSentinel emphasizes threshold-based behavioral detection to reduce false positives and improve adaptability.

The long-term goal of this project is to evolve the prototype into a scalable Network Detection & Response (NDR) platform.

3. System Architecture

CyberSentinel is built as a modular Python-based detection engine composed of three primary components:

1. Packet Capture Module
Responsible for capturing live network traffic using the Scapy library.

2. Detection Engine

Implements behavioral analysis logic for identifying suspicious activity such as ICMP flood and SYN scan attacks.

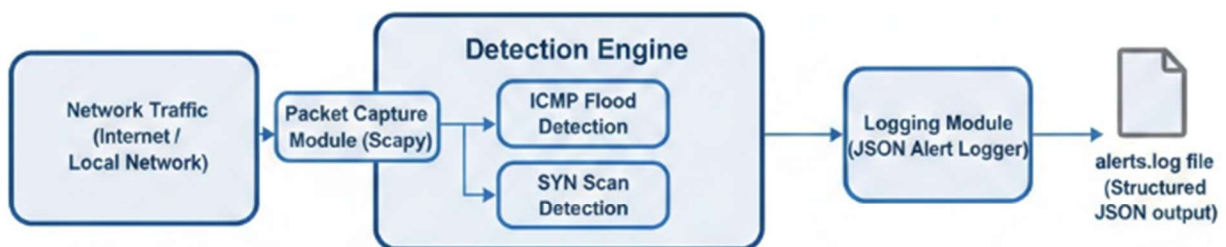
3. Logging Module

Generates structured JSON alerts and stores them in a local log file for future analysis or SIEM integration.

The system operates in real time by continuously analyzing captured packets and applying detection logic within a sliding time window.

This modular structure allows future integration of additional detection modules without modifying the core architecture.

CyberSentill – System Architecture Overview



The modular architecture allows future integration of additional detection modules such as brute-force detection, anomaly scoring, and automated response mechanisms.

4. Implementation Overview

CyberSentinel is implemented in Python and leverages the Scapy library for real-time packet capture and analysis.

The system operates continuously by monitoring live network traffic on a selected network interface. Captured packets are processed through a modular detection engine that applies protocol-specific behavioral logic.

4.1 Packet Capture

The packet capture process is handled using Scapy's `sniff()` function. The system listens on a specified network interface and filters IP-based traffic to ensure protocol-level analysis.

Each captured packet is forwarded to a central callback function responsible for protocol inspection and threat evaluation.

4.2 Modular Detection Functions

The detection engine is structured into separate functions for each attack type:

- `detect_icmp()` for ICMP flood detection
- `detect_syn()` for SYN scan detection

This modular design ensures that new detection modules can be added without modifying the core packet processing logic.

4.3 Sliding Time Window Mechanism

To identify abnormal behavior, the system implements a sliding time window approach. For each source IP address, timestamps of suspicious packets are stored in memory.

Packets outside the defined time window (5 seconds) are automatically discarded. If the number of suspicious packets exceeds a predefined threshold, the activity is classified as malicious.

This approach allows CyberSentinel to differentiate between normal traffic and attack patterns in real time.

5. Detection Methodology

CyberSentinel implements behavioral, threshold-based detection techniques to identify abnormal network activity in real time.

Rather than relying exclusively on signature matching, the system analyzes traffic patterns within a defined time window and triggers alerts when suspicious behavior exceeds predefined limits.

5.1 ICMP Flood Detection

ICMP flood detection is based on monitoring ICMP Echo Request packets (Type 8).

For each source IP address, the system tracks the number of ICMP Echo Requests received within a 5-second sliding time window.

Detection parameters:

- Protocol: ICMP
- Type: 8 (Echo Request)
- Time Window: 5 seconds
- Threshold: 20 packets
- Severity Level: HIGH
- Threat Score: 70

If the number of ICMP Echo Requests from a single source IP exceeds the defined threshold within the time window, the activity is classified as a potential ICMP flood attack.

This method effectively distinguishes between normal ping activity and high-volume flooding behavior, minimizing false positives.

[illegible]

6. Testing & Attack Simulation

To validate the effectiveness of CyberSentinel, controlled attack simulations were performed in a virtual lab environment using VirtualBox.

The lab consisted of:

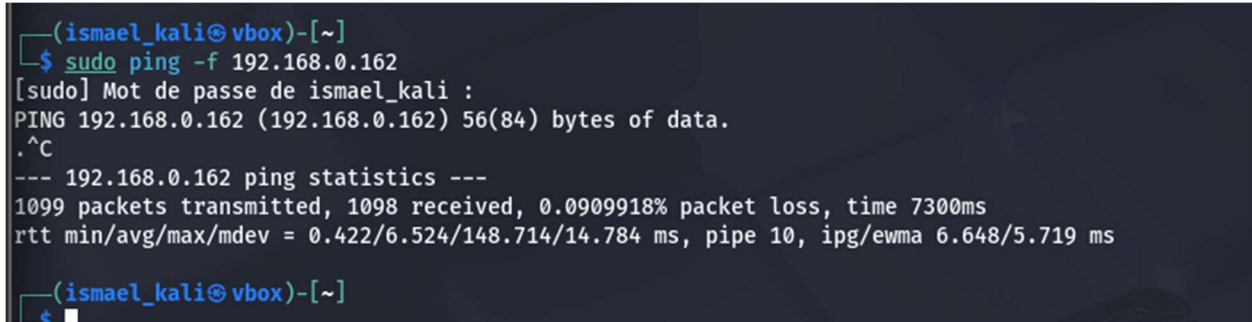
- Ubuntu Server (CyberSentinel IDS)
- Kali Linux (Attacker machine)
- Bridged network configuration

6.1 ICMP Flood Simulation

An ICMP flood attack was simulated from the Kali Linux machine using the following command: `sudo ping -f <target ip>`

This command generates a high volume of ICMP Echo Requests directed at the CyberSentinel host.

The detection engine successfully identified the abnormal packet rate and triggered a HIGH severity alert when the threshold was exceeded.

A terminal window with a dark background. The prompt is (ismael_kali@vbox)-[~]. The user enters \$ sudo ping -f 192.168.0.162. The terminal shows [sudo] Mot de passe de ismael_kali : followed by a series of dots. Then it shows PING 192.168.0.162 (192.168.0.162) 56(84) bytes of data. The user presses Ctrl-C (^C). The terminal shows --- 192.168.0.162 ping statistics ---. The statistics are: 1099 packets transmitted, 1098 received, 0.0909918% packet loss, time 7300ms. The round trip times are: rtt min/avg/max/mdev = 0.422/6.524/148.714/14.784 ms, pipe 10, ipg/ewma 6.648/5.719 ms. The prompt returns to (ismael_kali@vbox)-[~].

```
(ismael_kali@vbox)-[~]
$ sudo ping -f 192.168.0.162
[sudo] Mot de passe de ismael_kali :
PING 192.168.0.162 (192.168.0.162) 56(84) bytes of data.
.^C
--- 192.168.0.162 ping statistics ---
1099 packets transmitted, 1098 received, 0.0909918% packet loss, time 7300ms
rtt min/avg/max/mdev = 0.422/6.524/148.714/14.784 ms, pipe 10, ipg/ewma 6.648/5.719 ms
(ismael_kali@vbox)-[~]
```

During the ICMP flood simulation, the system generated multiple structured alerts in JSON format.

Each alert contains precise timestamp information and classification details, demonstrating real-time detection and logging capabilities.

The rapid sequence of alerts confirms that the threshold-based detection mechanism is functioning correctly under high packet rate conditions.

6.2 SYN Scan Simulation

A TCP SYN scan was performed from the Kali Linux machine using the following command:
`nmap -sS <target_ip>`

The `-sS` option performs a stealth SYN scan by sending TCP SYN packets to multiple ports without completing the full TCP handshake.

During the simulation, CyberSentinel detected an abnormal number of SYN packets within the defined 5-second time window.

Once the threshold was exceeded, the system generated a MEDIUM severity alert and logged the event in structured JSON format.

Figure 5 – SYN Scan Simulation from Kali Linux

```
(ismael_kali@vbox)-[~]  
$ nmap -sS 192.168.0.162  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-21 20:06 AST  
Nmap scan report for 192.168.0.162  
Host is up (0.0050s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
```

Figure 6 – SYN Scan Detection Alert on CyberSentinel

[illegible]

Figure 7 – Structured SYN_SCAN Log Entry


```
{
  "timestamp": "2026-02-22T00:06:42.426962",
  "src_ip": "192.168.0.135",
  "attack_type": "SYN_SCAN",
  "threat_score": 60,
  "severity": "MEDIUM"
}
{"timestamp": "2026-02-22T00:06:42.427483", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.431410", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.432181", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.432917", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.433777", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.438908", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.442624", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.444570", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.445567", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.447916", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.448643", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.449206", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.449904", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.452743", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.455789", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.456494", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.457579", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.458294", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.458851", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.459466", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.459980", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.465805", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.466408", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.466887", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
```

7. Results & Analysis

The testing phase demonstrated that CyberSentinel successfully differentiates between normal traffic and malicious behavior.

7.1 Normal Traffic Behavior

When standard ICMP requests were sent using a normal ping command: `ping <target_ip>`

```
(ismael_kali@vbox)-[~]
$ ping 192.168.0.162
PING 192.168.0.162 (192.168.0.162) 56(84) bytes of data.
64 bytes from 192.168.0.162: icmp_seq=1 ttl=64 time=1.66 ms
64 bytes from 192.168.0.162: icmp_seq=2 ttl=64 time=2.12 ms
64 bytes from 192.168.0.162: icmp_seq=3 ttl=64 time=1.87 ms
64 bytes from 192.168.0.162: icmp_seq=4 ttl=64 time=1.70 ms
64 bytes from 192.168.0.162: icmp_seq=5 ttl=64 time=3.36 ms
^C
--- 192.168.0.162 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4146ms
rtt min/avg/max/mdev = 1.660/2.142/3.358/0.629 ms

(ismael_kali@vbox)-[~]
```

The system did not generate any alerts.

```
ismael@ismael:~/cybersentinel$ sudo python3 main.py
[sudo] password for ismael:
[*] CyberSentinel running on enp0s3
```

This can be seen again in the logs; the last alert is the syn_scan performed with the nmap command.

```
{
  "timestamp": "2026-02-22T00:06:42.669054",
  "src_ip": "192.168.0.135",
  "attack_type": "SYN_SCAN",
  "threat_score": 60,
  "severity": "MEDIUM"
}
{"timestamp": "2026-02-22T00:06:42.673850", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.683076", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.688249", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.698275", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.708691", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.717930", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.723226", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.726611", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.730681", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.738522", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.743046", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T00:06:42.748398", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
ismael@ismael:~/cybersentinel$
```

This confirms that the threshold-based detection mechanism effectively prevents false positives during regular network activity.

7.2 ICMP Flood Detection Accuracy

During ICMP flood simulation using: `sudo ping -f <target_ip>`

CyberSentinel successfully detected excessive ICMP Echo Requests within the 5-second time window and generated HIGH severity alerts.

The detection occurred in real time, confirming the responsiveness of the behavioral monitoring logic.

```
(ismael_kali@vbox)-[~]
$ sudo ping -f 192.168.0.162
[sudo] Mot de passe de ismael_kali :
PING 192.168.0.162 (192.168.0.162) 56(84) bytes of data.
^C
--- 192.168.0.162 ping statistics ---
349 packets transmitted, 348 received, 0.286533% packet loss, time 882ms
rtt min/avg/max/mdev = 0.467/1.578/7.945/0.796 ms, pipe 2, ipg/ewma 2.535/1.457 ms
```

[illegible]

7.3 SYN Scan Detection Accuracy

During TCP SYN scan simulation using: `nmap -sS <target_ip>`

The system detected abnormal SYN packet rates and generated MEDIUM severity alerts.

This demonstrates that the sliding window mechanism effectively identifies reconnaissance behavior prior to exploitation attempts.

```
(ismael_kali@vbox)-[~]  
$ nmap -sS 192.168.0.162  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-21 21:22 AST  
Nmap scan report for 192.168.0.162  
Host is up (0.0042s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.61 seconds  
  
(ismael_kali@vbox)-[~]
```



```
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
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[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
[ALERT LOGGED] SYN_SCAN from 192.168.0.135
[⚠️ ALERT] Possible SYN Scan from 192.168.0.135
```

7.4 Logging Reliability

All detected attacks were recorded in structured JSON format within the alerts.log file.

The structured format ensures compatibility with future SIEM integration and automated log analysis tools.

```
{
  "timestamp": "2026-02-22T01:20:52.935488",
  "src_ip": "192.168.0.135",
  "attack_type": "ICMP_FLOOD",
  "threat_score": 70,
  "severity": "HIGH"
}
{"timestamp": "2026-02-22T01:20:52.937496", "src_ip": "192.168.0.135", "attack_type": "ICMP_FLOOD", "threat_score": 70, "severity": "HIGH"}
{"timestamp": "2026-02-22T01:20:52.942420", "src_ip": "192.168.0.135", "attack_type": "ICMP_FLOOD", "threat_score": 70, "severity": "HIGH"}
{"timestamp": "2026-02-22T01:20:52.944052", "src_ip": "192.168.0.135", "attack_type": "ICMP_FLOOD", "threat_score": 70, "severity": "HIGH"}
{"timestamp": "2026-02-22T01:20:52.947253", "src_ip": "192.168.0.135", "attack_type": "ICMP_FLOOD", "threat_score": 70, "severity": "HIGH"}
{"timestamp": "2026-02-22T01:20:52.949799", "src_ip": "192.168.0.135", "attack_type": "ICMP_FLOOD", "threat_score": 70, "severity": "HIGH"}
{"timestamp": "2026-02-22T01:20:52.957280", "src_ip": "192.168.0.135", "attack_type": "ICMP_FLOOD", "threat_score": 70, "severity": "HIGH"}
{"timestamp": "2026-02-22T01:22:56.914239", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.916718", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.917622", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.919384", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.920486", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.927025", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.929529", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.932223", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.932875", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
{"timestamp": "2026-02-22T01:22:56.933707", "src_ip": "192.168.0.135", "attack_type": "SYN_SCAN", "threat_score": 60, "severity": "MEDIUM"}
```

8. Automated Response Mechanism

CyberSentinel includes an automated response mechanism that dynamically blocks malicious IP addresses upon detection.

When an ICMP flood or SYN scan exceeds the configured threshold, the system automatically executes a Linux firewall rule using iptables to drop incoming traffic from the offending IP address.

The following command is applied programmatically: `iptables -A INPUT -s <malicious_ip> -j DROP`

This ensures that further attack traffic is immediately mitigated.

8.1 Firewall Rule Verification

During testing, the firewall rule was successfully added to the INPUT chain, as confirmed using: `sudo iptables -L`

```
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[^ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
^Cismael@ismael:~/cybersentinel$ ^C
ismael@ismael:~/cybersentinel$ ^C
ismael@ismael:~/cybersentinel$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP       all  --  192.168.0.135          anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```



```
ismael@ismael:~/cybersentinel$ sudo python3 main.py
[*] CyberSentinel running on enp0s3
[⚠️ ALERT] ICMP Flood detected from 192.168.0.135
[ALERT LOGGED] ICMP_FLOOD from 192.168.0.135
[🔒 BLOCKED] IP 192.168.0.135 has been blocked by firewall
```

9. Future Improvements

While CyberSentinel v1 successfully demonstrates real-time behavioral detection and automated mitigation, several enhancements are planned to evolve the system into a more advanced Network Detection & Response (NDR) platform.

9.1 Stateful Flow Tracking

Future versions will implement full 5-tuple flow tracking (source IP, destination IP, source port, destination port, protocol) to enable stateful inspection and session-based analysis.

This will allow more accurate detection of complex attack patterns.

9.2 SSH Brute-Force Detection

A module for detecting repeated failed SSH login attempts will be added.

This enhancement will allow detection of credential-based attacks in addition to network reconnaissance and flooding attacks.

9.3 Threat Scoring System

A cumulative threat scoring mechanism will be implemented to dynamically adjust severity levels based on multiple suspicious behaviors from the same IP address.

This will improve prioritization and reduce false positives.

9.4 Web-Based Monitoring Dashboard

A lightweight web dashboard will be developed using FastAPI and WebSocket to visualize alerts in real time and provide centralized monitoring capabilities.

9.5 Anomaly-Based Detection (Machine Learning)

Future iterations may integrate machine learning techniques such as Isolation Forest to detect traffic anomalies beyond fixed thresholds.

This would allow CyberSentinel to detect unknown or evolving attack patterns.

10. Conclusion

CyberSentinel v1 demonstrates the successful implementation of a lightweight behavioral Network Intrusion Detection and Response prototype.

The system is capable of:

- Capturing live network traffic in real time
- Detecting ICMP flood attacks
- Identifying TCP SYN scan reconnaissance activity
- Generating structured JSON security alerts
- Automatically blocking malicious IP addresses using firewall rules
- Preventing alert storms through suppression mechanisms

The project highlights the practical application of network protocol analysis, threshold-based behavioral detection, and automated mitigation strategies within a controlled lab environment.

Through attack simulation using Kali Linux, CyberSentinel demonstrated reliable real-time detection capabilities while maintaining stability during normal network activity.

Although currently implemented as a prototype, the modular architecture provides a strong foundation for future expansion into a more advanced Network Detection & Response platform.

This project reflects a strong understanding of network security fundamentals, defensive system design, and practical attack simulation methodologies.